

# DSA Day 1: Introduction to Data Structures

Book: Data Structures Through C in Depth - Ch1 §1.1-1.6 · pp 1-9 | Code in Python

## Overview

A data structure is a way to organise data so operations (insert/search/delete/traverse) are efficient. An ADT specifies WHAT operations do; the data structure specifies HOW.

## Key concepts to master

- Data type vs ADT vs data structure
- Primitive vs non-primitive; linear vs non-linear
- Algorithm: finite, definite, effective, has input/output
- Efficiency = time + space; measured with Big-O
- Best / worst / average case

## Python implementation

```
def growth(n):  
    # compare how fast terms grow  
    return {'n':n, 'nlogn':n*(n.bit_length()), 'n2':n*n, '2^n':1<<n}
```

## Complexity

Comparing terms as n grows:  $1 < \log n < n < n \log n < n^2 < 2^n < n!$

## Common pitfalls

- Confusing an ADT (stack) with its implementation (array/linked list)
- Assuming worst case always happens

## Practice

- Classify 5 operations by Big-O
- Write the ADT of a Stack in your own words

